Yu Sheng (Nicholas) Ng

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PROJECTS

Shock Resistant Building Model **Sydney, NSW**
DESN1000 | Overall course mark 90/100 (High Distinction) *2025*

- Evaluated 3 design concepts against 7 criteria within a 5-person team, incorporating a truss base from an alternative concept to address progressive-collapse risk.
- Modelled static loading and removal of 1 of 4 main columns (25%) in SPACE GASS to assess load redistribution through the truss base and X-bracing.
- Fabricated a 388 g balsa-wood prototype with teammates in the UNSW Makerspace, meeting the 400 g mass limit.

Steel Garage in Revit **Sydney, NSW**
Independent BIM and Structural Modelling Project *2026*

- Modelled a 6 × 9 m steel garage in Revit with three portal frames, roof and wall bracing, and six pad footings.
- Compared 3-frame and 4-frame layouts, weighing 4.5 m versus 3 m bay spacing against the extra columns, footings and connections.
- Produced an overview sheet in Revit with exterior and roof framing views and traced the intended wind load path from roof bracing to pad footings.

Randwick City Council x UNSW Case Competition **Sydney, NSW**
3rd Place out of 35+ Entries | Randwick Defence Land Redevelopment *2026*

- Led proposal development, directing the project, delegating team tasks and preparing the competition pitch.
- Developed a 10–12-year implementation plan covering approvals, site preparation and staged construction, incorporating a two-year buffer for delays and measures to meet emissions targets and protect native habitats.
- Defined five performance targets for the proposed precinct, including 70% social or affordable housing, 100% renewable electricity and 90% construction waste reuse or recycling.

Sumobots Competition **Sydney, NSW**
Mechanical Design and Fabrication *2025*

- Designed the mechanical model for an autonomous competition robot in Autodesk AutoCAD and translated the design into 3D-printed components.
- Fabricated and assembled the Sumobot in the UNSW Makerspace, collaborating with teammates responsible for Arduino programming to integrate the mechanical and electronic systems.

EXPERIENCE

Year 12 Formal **Sydney, NSW**
Communications and Administration Lead *2024*

- Managed emails to about 150 classmates for a student-run formal with around 250 attendees.
- Coordinated promotion for a formal with about $27,000 in ticket sales and a 10% rise in attendance.
- Consolidated Google Forms responses in Excel so the organising team could track attendance and follow up.
- Built a budget of about $25,000, breaking down costs for the venue, vendors and other event purchases.

Sydney Boys High School **Sydney, NSW**
Prefect *2023 - 2024*

- Delivered 10+ student initiatives as part of a 28-member prefect team.
- Organised an interschool event attended by at least 90 students from 30 schools and educational workshops for approximately 100 students.

SKILLS AND CERTIFICATIONS

White Card: general construction induction card | **Microsoft Excel:** calculations, laboratory data analysis, Gantt charts and response tracking | **Autodesk Revit**: BIM and structural modelling | **Autodesk AutoCAD:** 3D modelling | **SPACE GASS:** structural analysis | **Fabrication:** 3D printing and prototype assembly

EDUCATION

University of New South Wales **Sydney, NSW**
Bachelor of Engineering (Civil) (Honours) | WAM: 64 *2025 - Expected graduation: Dec 2028*

Sydney Boys High School **Sydney, NSW**
High School Certificate *2024*